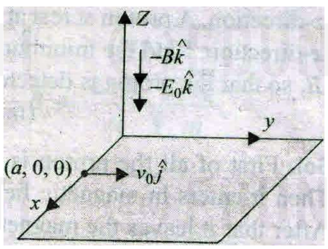
SUBJECTIVE QUESTIONS

1. A positively charged particle of mass m and charge q is projected on a rough horizontal x-y plane surface with z-axis in the vertically upward direction. Both electric and magnetic fields are acting in the region and given by $\vec{E} = -E_0\hat{k}$ and $\vec{B} = -B_0\hat{k}$, respectively. The particle enters into the field at (a_0, 0, 0) with velocity $\vec{v} = v_0\hat{j}$. The particle starts moving in some curved path on the plane. If the coefficient of friction between the particle and the plane is μ, calculate the

(a) time when the particle will come to rest

(b) distance travelled by the particle when it comes to rest



Solution Electrostatic force on the particle is in downward direction. Hence, the total normal force on the particle is

$$N = mg + qE_0$$

So, frictional force, $f = \mu N = \mu(mg + qE_0)$

Its acceleration in the tangential direction is constant and is given by

$$a = \frac{-\mu}{m}(mg + qE_0)$$

The particle moves in a spiral. Due to decreasing speed, it finally comes to rest in time

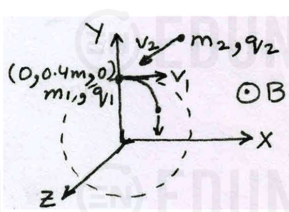
$$t = \frac{v-u}{a} = \frac{0 - v_0}{-\mu(mg+qE_0)/m} = \boxed{\frac{mv_0}{\mu(mg+qE_0)}}$$

The distance travelled before coming to rest is

$$s = \frac{v^2 - u^2}{2a} = \frac{0^2 - v_0^2}{-2\mu(mg+qE_0)/m} = \boxed{\frac{mv_0^2}{2\mu(mg+qE_0)}}$$

2. A positively charged particle having charge q_1 = 1 C and mass m_1 = 40 gm is revolving along a circle of radius r_1 = 40 cm with velocity v_1 = 5 m s^{-1} in a uniform magnetic field with center of circle at origin O of a three-dimensional system. At t = 0, the particle was at (0, 0.4m, 0) and velocity was directed along positive x direction. Another particle having charge q_2 = 1 C and mass m_2 = 10 g moving uniformly parallel to positive z-direction with velocity $v_2 = 40/\pi\, m s^{-1}$ collides with revolving particle at t = 0 and gets stuck to it. Neglecting gravitational force and coulomb force, calculate x-, y- and z- coordinates of the combined particle at $t = (\pi/40)$s.

Solution



Here, $q_1 = 1$C, $m_1 = 40$g, $v_1 = 5$m/s, $r_1 = 40$cm, $q_2 = 1$C, $m_2 = 10$g, and $v_2 = (40/\pi)$ m/s

Before collision, particle 1 is revolving in a circle. So,

$$q_1 v_1 B = \frac{m v_1^2}{r_1} \qquad \therefore\ B = \frac{m_1 v_1}{q_1 r_1} = \frac{0.04\times 5}{1\times 0.4} = 0.5T.$$

After collision, the charge of the combined particle is

$$q = q_1 + q_2 = 1 + 1 = 2C$$

and its velocity by conservation of momentum is

$$\vec{v} = \frac{m_1\vec{v_1} + m_2\vec{v_2}}{m_1+m_2} = \frac{0.04\times 5\hat{i} + 0.01\times(40/\pi)\hat{k}}{0.04+0.01} = 4\hat{i} + \frac{8}{\pi}\hat{k}$$

Its path is a helix. Its projection in xy plane is a circle whose radius is

$$r = \frac{(m_1+m_2)v_x}{qB} = \frac{(0.04+0.01)\times 4}{2\times 0.5} = 0.2m$$

and time period is

$$T = \frac{2\pi(m_1+m_2)}{qB} = \frac{2\pi(0.04+0.01)}{2\times 0.5} = \frac{\pi}{10}s$$

In time $t = (\pi/40)s = T/4$, the combined particle has rotated by $90°$ in a circle of radius 0.2m. Hence, its x and y coordinates are respectively

$$x = 0.2\,\text{m} \quad \text{and} \quad y = 0.4 - 0.2 = 0.2\,\text{m}.$$

Its z coordinate is

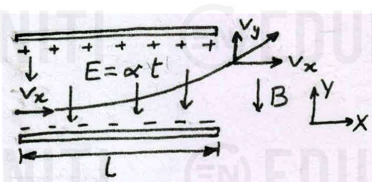
$$z = v_z t = \frac{8}{\pi}\times\frac{\pi}{40} = 0.2\,\text{m}$$

Therefore, in time $t = (\pi/40)$s, its coordinates are

$$\boxed{(0.2\,\text{m},\ 0.2\,\text{m},\ 0.2\,\text{m})}$$

3. An electron of charge '- e' and mass m accelerated by a potential difference V first enters into a uniform electric field of a parallel plate capacitor whose plates extend over a length l in the direction of initial velocity. The electric field is normal to the direction of initial velocity and its strength varies with time as $E = \alpha t$, where α is a constant. Then the electron enters into a uniform magnetic field of induction B. Direction of magnetic field is same as that of the electric field. Calculate pitch of helical path traced by the electron in the magnetic field. (Mass of electron, $m = 9\times 10^{-31} kg$)

Solution



When the electron is accelerated through a potential difference V, it acquires kinetic energy equal to eV.

$$\Rightarrow \frac{1}{2}mv_x^2 = eV$$

The electron enters the capacitor with velocity v_x. This velocity inside the capacitor along x-direction does not change. So, the time taken by electron to cross the capacitor is

$$t_0 = l / v_x .$$

During this time, it is subjected to acceleration along y-axis given by

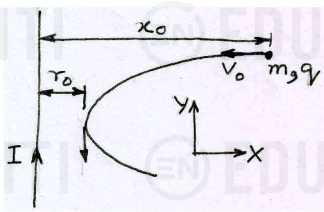
$$a_y = \frac{eE}{m} = \frac{e}{m}\alpha t \quad \Rightarrow \int_0^{v_y} dv_y = \frac{e\alpha}{m}\int_0^{t_0} t\,dt$$

$$\Rightarrow \; v_y = \frac{e\alpha}{2m}t_0^2 = \frac{e\alpha}{2m}\frac{l^2}{v_x^2} = \frac{e\alpha l^2}{4eV} = \frac{\alpha l^2}{4V}$$

The pitch of the helical path after it enters the magnetic field is

$$p = \frac{2\pi m}{eB}v_y = \frac{2\pi m}{eB}\times\frac{\alpha l^2}{4V} = \boxed{\frac{\pi m\alpha l^2}{4eBV}}$$

4.



A long straight wire carries a current I. A particle having a positive charge q and mass m, kept at distance x_0 from the wire is projected towards it with speed v_0. Find the closest distance of approach r_0 of charged particle to the wire.

Solution Let the instantaneous velocity of the particle be $\vec{v} = v_x\hat{i} + v_y\hat{j}$ when it is at distance x from the wire.

The magnetic field due to wire at this location is

$$\vec{B} = \frac{-\mu_0 I}{2\pi x}\hat{k}$$

The magnetic force on the particle is

$$\vec{F} = q(\vec{v}\times\vec{B}) = -q(v_x\hat{i} + v_y\hat{j})\times\left(\frac{\mu_0 I}{2\pi x}\hat{k}\right)$$

$$\Rightarrow m\left(\frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j}\right) = \frac{\mu_0 qI}{2\pi x}(-v_y\hat{i} + v_x\hat{j})$$

$$\Rightarrow \frac{dv_y}{dt} = \frac{\mu_0 qI}{2\pi mx}v_x = \frac{\mu_0 qI}{2\pi mx}\frac{dx}{dt} \qquad \Rightarrow dv_y = \frac{\mu_0 qI}{2\pi mx}\frac{dx}{x}$$

We know that the speed of the charged particle in a magnetic field do not change. Hence, at the minimum distance r_0,

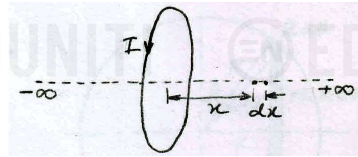
$$v_y = -v_0$$

$$\Rightarrow \int_0^{-v_0} dv_y = \frac{\mu_0 qI}{2\pi m}\int_{x_0}^{r_0}\frac{dx}{x} \qquad \Rightarrow -v_0 = \frac{\mu_0 qI}{2\pi m}\ln\frac{r_0}{x_0}$$

$$\therefore \boxed{r_0 = x_0 l^{-2\pi m v_0/\mu_0 qI}}$$

5. A current I flows along a round loop. Find the integral $\int \vec{B}.d\vec{r}$ along the axis of the loop within the range from $-\infty$ to $+\infty$. Explain the result obtained

Solution



Magnetic field due to loop along the axis at a distance x from centre is

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

So, $$\int_{-\infty}^{\infty}\vec{B}\cdot\vec{dr} = \frac{\mu_0 I R^2}{2}\int_{-\infty}^{\infty}\frac{dx}{(R^2 + x^2)^{3/2}}$$

Take $x = R\tan\theta \qquad \Rightarrow dx = R\sec^2\theta\, d\theta$

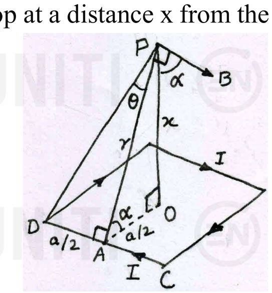
$$\therefore \int_{-\infty}^{\infty}\vec{B}\cdot\vec{dr} = \frac{\mu_0 I R^2}{2}\int_{-\pi/2}^{\pi/2}\frac{R\sec^2\theta}{R^3\sec^3\theta}d\theta = \frac{\mu_0 I}{2}\int_{-\pi/2}^{\pi/2}\cos\theta\, d\theta$$

$$= \boxed{\mu_0 I}$$

The term on the LHS can be considered as line integral of B over a closed path (a straight line from $-\infty$ to $+\infty$ and a semicircle of infinite radius). Note that at infinite radius, magnetic field is zero and hence over this length the integral of B is zero. So, as per Ampere's law, LHS must be equal to μ_0 times current enclosed by the path which is I. Hence, the result.

6. A square loop of wire, edge length a, carries a current I. Compute the magnitude of the magnetic field produced at a point on the axis of the loop at a distance x from the centre.

Solution



In ΔOAP, $\quad r = \sqrt{x^2 + (a/2)^2}$

In ΔAPD, $\quad \sin\theta = \frac{a/2}{\sqrt{r^2 + (a/2)^2}} = \frac{a}{\sqrt{4x^2 + 2a^2}}$

Magnetic field at P due to wire CD is

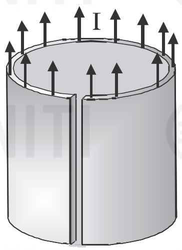
$$B = \frac{\mu_0 I}{4\pi r}(\sin\theta + \sin\theta) = \frac{\mu_0 I\sin\theta}{2\pi r}$$

Its component along PO is $B\cos\alpha$. The component perpendicular to PO due to all four wires of square gets cancelled. The net magnetic field due to all four wires is

$$B_{net} = 4B\cos\alpha = 4\left(\frac{\mu_0 I\sin\theta}{2\pi r}\right)\frac{a}{2r} = \frac{\mu_0 Ia}{\pi r^2}\sin\theta$$

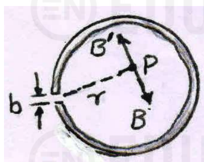
$$= \boxed{\frac{4\mu_0 Ia^2}{\pi\left(4x^2 + a^2\right)\sqrt{4x^2 + 2a^2}}}$$

7.



A current I flows along a thin walled tube of radius R with a long longitudinal slit of width b (<<R). What is the magnitude of magnetic field induction at a point inside the tube at a distance r from the slit?

Solution



We know that if there was no slit, the magnetic field inside the tube would have been zero. To find the magnetic field at point P, we shall use superposion principle.

The current per unit length of circumference is

$$\frac{I}{2\pi R - b}$$

If the slit was not there, the current through the slit would have been

$$I' = \frac{Ib}{2\pi R - b}$$

Let the magnetic field at point P due to the current I in the tube be B and due to the current I' through the slit be B'.

Then, by superposition principle,

$$\vec{B} + \vec{B'} = 0$$

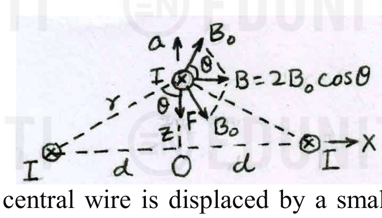
$$\therefore\ |\vec{B}| = |\vec{B'}| = \frac{\mu_0 I'}{2\pi r} = \boxed{\frac{\mu_0 Ib}{2\pi r(2\pi R - b)}}$$

8. Three infinitely long wires, each carrying current I in the same direction, are in the x-y plane of a gravity free space. The central wire is along the y-axis while the other two are along $x = \pm d$.

(a) If the central wire is displaced along the z-direction by a small amount and released, show that it will execute simple harmonic motion. If the linear mass density of the wire is λ, find the time period of oscillation.

Solution

(a)



When the central wire is displaced by a small distance z along z-zxis, the net magnetic field at its location due to other two wires is

$$B = 2B_0 \cos\theta = 2 \times \frac{\mu_0 I}{2\pi r} \times \frac{z}{r} = \frac{\mu_0 Iz}{\pi(d^2 + z^2)} \approx \frac{\mu_0 I z}{\pi d^2}$$

Note that for small z,

$$d^2 + z^2 \approx d^2$$

The restoring force on the central wire of length L is towards the equilibrium position O given by

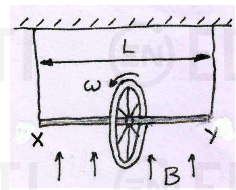
$$F = ILB = IL \times \frac{\mu_0 I z}{\pi d^2} = \frac{\mu_0 I^2 L z}{\pi d^2}$$

$$\Rightarrow\ (\lambda L)a = \frac{\mu_0 I^2 L z}{\pi d^2} \quad \Rightarrow\ a = -\frac{\mu_0 I^2}{\pi \lambda d^2} z$$

As $a \propto z$ with a negative sign, the wire will execute simple harmonic motion.

Now, $\omega^2 = \dfrac{\mu_0 I^2}{\pi \lambda d^2}$ $\qquad \therefore\ \boxed{T = \dfrac{2\pi}{\omega} = \dfrac{2\pi d}{I}\sqrt{\dfrac{\pi\lambda}{\mu_0}}}$

9.



A ring of radius R having uniformly distributed charge Q is mounted at the centre of a rod XY of length L and suspended by two identical strings. The tension in strings in equilibrium is T_0. Now a vertical magnetic field is switched on and ring is rotated at constant angular velocity ω. Find the maximum ω with which the ring can be rotated if the strings can withstand a maximum tension of $3T_0/2$.

Solution Let M be the total mass of the rod and the ring. Its centre of mass is at the centre of the rod. Initially in equilibrium,

$$2T_0 = Mg$$

When the ring rotates with angular velocity ω, the current in the ring is

$$i = \frac{Q}{T} = \frac{Q\omega}{2\pi}$$

Its magnetic moment is towards right given by

$$M = iA = \frac{Q\omega}{2\pi} \times \pi R^2 = \frac{Q\omega R^2}{2}$$

The torque experienced by the ring is $\vec{\tau} = \vec{M} \times \vec{B}$.

Its direction is anticlockwise and magnitude is

$$\tau = \frac{Q\omega R^2}{2} B.$$

Due to this torque, the tension in the left string increases and that in the right string decreases. Let T be the tension in the left string. Taking torque about point Y, we have

$$TL = Mg\frac{L}{2} + \frac{Q\omega R^2 B}{2} \qquad \Rightarrow\ \frac{Q\omega R^2 B}{2} = TL - Mg\frac{L}{2}$$

where $T_{max} = 3T_0/2$ and $Mg = 2T_0$

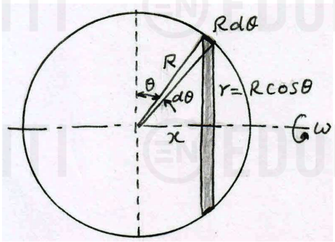
$$\therefore\ \omega_{max} = \frac{2}{QR^2 B}\left(\frac{3T_0}{2}L - 2T_0 \times \frac{L}{2}\right) = \boxed{\frac{T_0 L}{QR^2 B}}$$

10. A sphere of radius R, uniformly charged with the surface charge density σ, rotates around the axis passing through its center at an angular velocity ω.

(a) Find the magnetic field at the center of the rotating sphere.

(b) Find its magnetic moment.

Solution



(a) Consider a small circular ring as shown in figure. Its radius is $r = R\cos\theta$ and the distance of its centre from the centre of sphere is $x = R\sin\theta$.

Its surface area is

$$dA = (2\pi R\cos\theta)Rd\theta = 2\pi R^2\cos\theta d\theta.$$

The charge on this ring is

$$dq = \sigma dA = 2\pi R^2\sigma\cos\theta d\theta$$

The current associated with this charge is

$$dI = \frac{dq}{T} = \frac{\omega dq}{2\pi} = \omega R^2\sigma\cos\theta d\theta$$

The magnetic field due to this ring at the centre of sphere is

$$dB = \frac{\mu_0 (dI)r^2}{2(x^2+r^2)^{3/2}} = \frac{\mu_0}{2}\times\frac{\omega R^2\sigma\cos\theta d\theta}{R^3}\times R^2\cos^2\theta$$

$$= \frac{\mu_0}{2}\sigma\omega R\cos^3\theta\, d\theta$$

$$\therefore\ B = \frac{\mu_0}{8}\sigma\omega R\int_{-\pi/2}^{\pi/2}(\cos 3\theta + 3\cos\theta)d\theta$$

$$= \frac{\mu_0\sigma\omega R}{8}\left[\frac{\sin 3\theta}{3} + 3\sin\theta\right]_{-\pi/2}^{\pi/2}$$

$$= \frac{\mu_0\sigma\omega R}{8}\left(-\frac{2}{3}+6\right) = \boxed{\frac{2}{3}\mu_0\sigma\omega R}$$

(b) The magnetic moment M of this sphere can be compared to the angular momentum L of a thin hollow sphere of same radius R and rotating with the same angular velocity ω.

$$\therefore\ M = \frac{q}{2m}\times L = \frac{\sigma\times 4\pi R^2}{2m}\times\frac{2}{3}mR^2\omega = \boxed{\frac{4}{3}\pi\sigma R^4\omega}$$

MORE THAN ONE CORRECT QUESTIONS

11. A positively charged particle with velocity $\vec{v} = x\hat{i} + y\hat{j}$ moves in a magnetic field $\vec{B} = y\hat{i} + x\hat{j}$. The force acting on the particle has magnitude F. Which of the following statement(s) is/are correct?

(a) If $x = y$, no force will act on charged particle

(b) If $x > y$, $F \propto (x^2 - y^2)$

(c) If $x > y$, force will act along z-axis

(d) If $y > x$, force will act along x-axis

11.(a,b,c) $\vec{F} = q\vec{v}\times\vec{B} = q(x\hat{i}+y\hat{j})\times(y\hat{i}+x\hat{j}) = q(x^2-y^2)\hat{k}$

Here, q is positive. Therefore,

if $x = y$, $F = 0$

if $x > y$, F is along positive z-axis and $F \propto (x^2 - y^2)$

if $y > x$, F is along negative z-axis

12. A particle of specific charge $q/m = \pi\times 10^{10}\ \text{C kg}^{-1}$ is projected from the origin along the positive x–axis with a velocity of $10^5\ \text{ms}^{-1}$ in a uniform magnetic field $\vec{B} = -2\times 10^{-3}\hat{k}$ tesla. Which of the following statements is/are correct?

(a) The centre of the circle lies on the y–axis

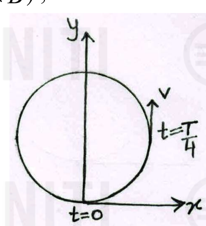
(b) The time period of revolution is 10^{-7} s.

(c) The radius of the circular path is $5/\pi$ mm

(d) The velocity of the particle at $t = 0.25\times 10^{-7}$ s is $10^5\hat{j}$ m/s

12.(a,b,c,d) $\frac{q}{m} = \pi\times 10^{10}\ \text{Ckg}^{-1}$, $\vec{v} = 10^5\hat{i}$ m/s and $\vec{B} = -2\times 10^{-3}\hat{k}$ T

Using $\vec{F} = q(\vec{v}\times\vec{B})$, the force acts along y-axis.



Thus centre of circle lies on y-axis.

$\therefore$ (a) is correct

Time period,

$$T = \frac{2\pi m}{qB} = \frac{2\pi m}{B\times(q/m)} = \frac{2\pi}{2\times 10^{-3}\times\pi\times 10^{10}} = 10^{-7}\text{ s}$$

$\therefore$ (b) is correct

Radius of path,

$$r = \frac{mv}{qB} = \frac{v}{(q/m)B} = \frac{10^5}{\pi\times 10^{10}\times 2\times 10^{-3}} = \frac{5}{\pi}\times 10^{-3}\text{ m} = \frac{5}{\pi}\text{mm}$$

$\therefore$ (c) is correct

At $t = 0.25\times 10^{-7}\text{ s} = T/4$, $\vec{v} = 10^5\hat{j}$ m/s

$\therefore$ (d) is correct

13. Velocity and acceleration vector of a charged particle moving in a uniform magnetic field at some instant are $\vec{v} = 3\hat{i} + 4\hat{j}$ and $\vec{a} = 2\hat{i} + x\hat{j}$. Select the correct options

(a) $x = -1.5$ (b) $x = 3$

(c) Magnetic field is along z-direction

(d) Kinetic energy of the particle is constant

13.(a,d) The force due to magnetic field is always perpendicular to the velocity. So, $\vec{a}\perp\vec{v}$ and $\vec{a}\cdot\vec{v} = 0$

$\Rightarrow\ (2\hat{i}+x\hat{j})\cdot(3\hat{i}+4\hat{j}) = 0 \qquad \Rightarrow\ 6 + 4x = 0$

$\Rightarrow\ x = -1.5 \qquad \therefore$ (a) is correct and (b) is incorrect

The kinetic energy of a particle in magnetic field is always constant. $\therefore$ (d) is correct.

Let $\vec{B} = a\hat{i} + b\hat{j} + c\hat{k}$

Then $m\vec{a} = q\vec{v} \times \vec{B}$

or $m(2\hat{i} + x\hat{j}) = q(3\hat{i} + 4\hat{j}) \times (a\hat{i} + b\hat{j} + c\hat{k})$

$\Rightarrow \quad 2\hat{i} + x\hat{j} = \frac{q}{m}[4c\hat{i} - 3c\hat{j} + (3b - 4a)\hat{k}]$

$\Rightarrow \quad c \neq 0. \qquad \therefore$ (c) is incorrect.

14. A charged particle P leaves the origin with speed $v = v_0$ at some inclination with the x-axis. There is a uniform magnetic field B along the x-axis. P strikes a fixed target T on the x-axis for a minimum value of $B = B_0$. P will also strike T if

(a) $B = 2B_0, v = 2v_0$ (b) $B = 2B_0, v = v_0$

(c) $B = B_0, v = 2v_0$ (d) $B = B_0/2, v = 2v_0$

14.(a, b) The particle P leaves from origin, target T lies on x-axis and the magnetic field B is also along x-axis. So, the distance x_0 (between origin and T) must be an integral multiple of pitch p.

$$\Rightarrow \quad x_0 = n \times \frac{2\pi m}{qB} \times v\cos\theta \text{ or } B = n\left(\frac{2\pi m v\cos\theta}{qx_0}\right)$$

The minimum value of $B = B_0$ is when $n = 1$ and $v = v_0$

$$\Rightarrow \quad \frac{2\pi m\cos\theta}{qx_0} = \frac{B}{nv} = \frac{B_0}{v_0} \text{ or } n = \frac{B}{B_0} \times \frac{v_0}{v}$$

P will also strike T if $n \geq 1$. This is satisfied by options (a) and (b).

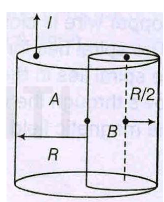
15. From a cylinder of radius R, a cylinder of radius $R/2$ is removed as shown in the figure. Current flowing in the remaining cylinder is I. Magnetic field strength is

(a) zero at point A

(b) zero at point B

(c) $\frac{\mu_0 I}{3\pi R}$ at point A

(d) $\frac{\mu_0 I}{3\pi R}$ at point B



15.(c,d) The given cyclinder can be considered as a combination of two solid cylinders; bigger one of radius R carrying current $4I/3$ and smaller one of radius $R/2$ carrying current $I/3$ in opposite direction.

$$\therefore \quad B_A = (B_A)_{big} + (B_A)_{small} = 0 + \frac{\mu_0(I/3)}{2\pi(R/2)} = \frac{\mu_0 I}{3\pi R} \text{ and}$$

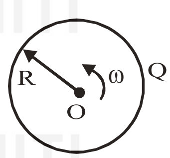
$$B_B = (B_B)_{big} + (B_A)_{small} = \frac{\mu_0(4I/3)}{2\pi R^2} \times \frac{R}{2} + 0 = \frac{\mu_0 I}{3\pi R}$$

16. A nonconducting disc having uniform positive charge Q, is rotating about its axis with uniform angular velocity ω. The magnetic field at the centre of the disc is

(a) directed outward

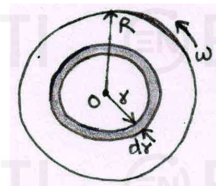
(b) having magnitude $\frac{\mu_0 Q\omega}{4\pi R}$

(c) directed inwards



(d) having magnitude $\frac{\mu_0 Q\omega}{2\pi R}$

16.(a, d) Consider an elemental ring of radius r and thickness dr. The charge on it is



$$dQ = \frac{Q}{\pi R^2} \times 2\pi r dr = \frac{2Q}{R^2} r dr$$

The current associated with it (due to rotation) is

$$dI = \frac{dQ}{T} = \frac{\omega}{2\pi} \times \frac{2Q}{R^2} r dr$$

Magnetic field at centre (due to this current) is

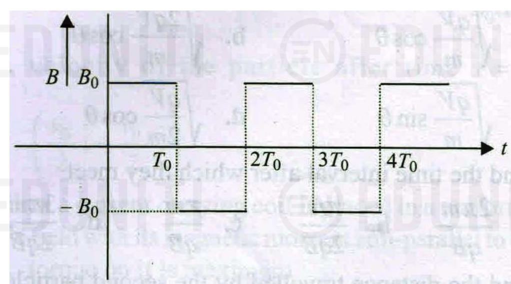
$$dB = \frac{\mu_0 dI}{2r} = \frac{\mu_0}{2r} \times \frac{Q\omega r dr}{\pi R^2} \quad \therefore \quad B = \frac{\mu_0 Q\omega}{2\pi R^2}\int_0^R dr = \frac{\mu_0 Q\omega}{2\pi R}$$

As the direction of current is anticlockwise, the magnetic field at O is directed outwards.

PARAGRAPH TYPE QUESTIONS

Paragraph for Question Nos. 17 to 18

In a region, the magnetic field along x-axis changes according to the graph given in fig.



The time period, radius and pitch of the helical path are T_0, R_0 and $3R_0$ respectively. The particle is projected at an angle θ_0 with the positive x-axis toward positive y-axis in x-y plane.

17. The coordinates of the particle at $t = 3T_0/2$ are

(a) $(9R_0/2, 0, -2R_0)$ (b) $(3R_0, 0, -R_0)$

(c) $(9R_0/2, 0, 2R_0)$ (d) $(3R_0, 0, R_0)$

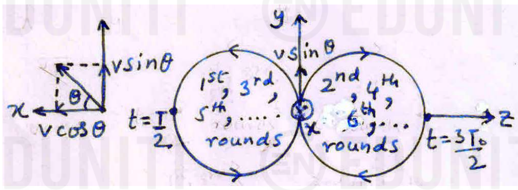
18. What is the maximum possible distance between any two positions during a time interval of one time period,. i.e., during $t_0 \leq t \leq t_0 + T_0$?

(a) $3R_0$ (b) $4R_0$

(c) $5R_0$ (d) $6R_0$

Paragraph (Solution 17 to 18)

17(c) The projection of the helical path on yz plane is a circle which lies on the $-z$ side in 1st, 3rd, 5th, ... rounds and on the $+z$ side in 2nd, 4th, 6th, ... rounds.



At $t = 3T_0/2$, the particle has made one and a half rounds. So, its x-coordinate is equal to one and a half pitch (i.e., $9R_0/2$), y-coordinate is zero and z-coordinate is $2R_0$.

18(c) In a time interval of one time period, the maximum distance will be when the z-coordinates at two positions are $-2R_0$ and $+2R_0$. Let us consider these two positions when $t = T/2$ and $t = 3T/2$. The coordinates of the two particle at these two positions are respectively $(3R_0/2, 0, -2R_0)$ and $(9R_0/2, 0, 2R_0)$. The distance between these two positions is $\sqrt{(3R_0)^2 + 0^2 + (4R_0)^2} = 5R_0$

Paragraph for Question Nos. 19 to 21

In a certain region of space, there exists a uniform and constant electric field of magnitude E along the negative y-axis of a coordinate system. A charged particle of mass m and charge q ($q > 0$) is projected from the origin with speed $2v$ at an angle of 60^o with the positive x-axis in x-y plane. When the x-coordinate of particle becomes $\sqrt{3}\,mv^2/qE$, a uniform and constant magnetic field of strength B is also switched on along positive y-axis. Let $t = 0$ at this instant.

19. Velocity of the particle just before the magnetic field is switched on is

(a) $v\hat{i}$ (b) $v\hat{i} + \dfrac{\sqrt{3}v}{2}\hat{j}$

(c) $v\hat{i} - \dfrac{\sqrt{3}v}{2}\hat{j}$ (d) $2v\hat{i} - \dfrac{\sqrt{3}v}{2}\hat{j}$

20. x-coordinate of the particle as a function of time after the magnetic field is switched on is

(a) $\dfrac{\sqrt{3}mv^2}{qE} - \dfrac{mv}{qB}\sin\left(\dfrac{qB}{m}t\right)$ (b) $\dfrac{\sqrt{3}mv^2}{qE} + \dfrac{mv}{qB}\sin\left(\dfrac{qB}{m}t\right)$

(c) $\dfrac{\sqrt{3}mv^2}{qE} - \dfrac{mv}{qB}\cos\left(\dfrac{qB}{m}t\right)$ (d) $\dfrac{\sqrt{3}mv^2}{qE} + \dfrac{mv}{qB}\cos\left(\dfrac{qB}{m}t\right)$

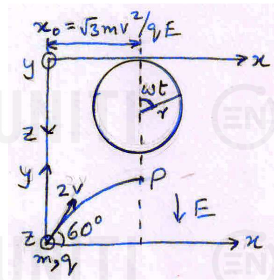
21. z-coordinate of the particle as a function of time after the magnetic field is switched on is

(a) $\dfrac{mv}{qB}\left[1-\cos\left(\dfrac{qB}{m}t\right)\right]$ (b) $-\dfrac{mv}{qB}\left[1-\sin\left(\dfrac{qB}{m}t\right)\right]$

(c) $-\dfrac{mv}{qB}\left[1-\cos\left(\dfrac{qB}{m}t\right)\right]$ (d) $\dfrac{mv}{qB}\left[1-\sin\left(\dfrac{qB}{m}t\right)\right]$

Paragraph (Solution 19 to 21)

19(a), 20(b), 21(a)



$v_x = 2v\cos 60^o = v$ and $v_y = 2v\sin 60^o = \sqrt{3}\,v$

Time taken to reach point P is

$$t_0 = \frac{x}{v_x} = \frac{\sqrt{3}\,mv^2}{qE} \times \frac{1}{v} = \frac{\sqrt{3}\,mv}{qE}$$

$$v_y = u_y + a_y t_0 = \sqrt{3}\,v - \frac{qE}{m} \times \frac{\sqrt{3}\,mv}{qE} = 0$$

Therefore, at point P (i.e., just before the magnetic field is switched on), the velocity of the particle is $v\hat{i}$

After the magnetic field is switched on, the partical moves in helical path with increasing pitch along $-y$ direction. Its projection in xy plane will be a circle whose radius and angular velocities are respectively

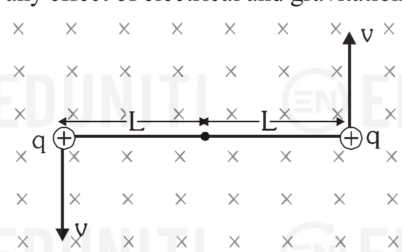
$$r = \frac{mv}{qB} \text{ and } \omega = \frac{v}{r} = \frac{qB}{m}$$

$$\therefore\quad x = x_0 + r\sin\omega t = \frac{\sqrt{3}\,mv^2}{qE} + \frac{mv}{qB}\sin\left(\frac{qB}{m}t\right)$$

$$\text{and}\quad z = r - r\cos\omega t = \frac{mv}{qB}\left[1 - \cos\left(\frac{qB}{m}t\right)\right]$$

Paragraph for Question Nos. 22 to 24

Two charged particles each of mass 'm', carrying charge $+q$ and connected with each other by a massless inextensible string of length $2L$ are describing circular path in the plane of paper. The speed of each particle is $v = qB_0L/m$ (where B_0 is constant) about their centre of mass. A uniform magnetic field $\vec{B}$ exists into the plane of paper as shown in figure. Neglect any effect of electrical and gravitational forces.



22. The magnitude of the magnetic field such that no tension is developed in the string will be

(a) $B_0/2$ (b) B_0

(c) $2B_0$ (d) 0

23. If the actual magnitude of magnetic field is half of that calculated above, then tension in the string will be

(a) $\dfrac{3}{4}\dfrac{q^2B_0^2L}{m}$ (b) zero

(c) $\frac{q^2B_0^2L}{2m}$ (d) $\frac{2q^2B_0^2L}{m}$

24. Given that the string breaks when the tension is $T=\frac{3}{4}\frac{q^2B_0^2L}{m}$. Now if the magnetic field is reduced to such a value that the string just breaks, then find the maximum separation between the two particles during their motion

(a) $16L$ (b) $4L$

(c) $14L$ (d) $2L$

Paragraph (Solution 22 to 24)

22.(b) $T+qvB=\frac{mv^2}{L}$

If $T=0$, then $qvB=\frac{mv^2}{L}$

$\therefore\ B=\frac{mv}{qL}=\frac{m}{qL}\times\frac{qB_0L}{m}=B_0$

23.(c) If $B=B_0/2$, then

$$T=\frac{mv^2}{L}-qvB=\frac{m}{L}\times\left(\frac{qB_0L}{m}\right)^2-q\left(\frac{qB_0L}{m}\right)\frac{B_0}{2}$$

$$=\frac{q^2B_0^2L}{m}-\frac{q^2B_0^2L}{2m}=\frac{q^2B_0^2L}{2m}$$

24.(c) From $T+qvB=mv^2/L$, we have

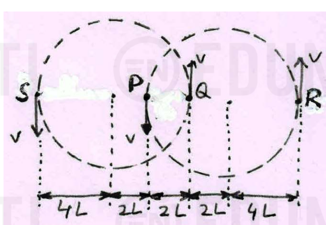
$$B=\frac{mv}{qL}-\frac{T}{qv}=\frac{m}{qL}\left(\frac{qB_0L}{m}\right)-\frac{1}{q}\times\frac{3}{4}\frac{q^2B_0^2L}{m}\times\frac{m}{qB_0L}$$

$$=B_0-\frac{3}{4}B_0=\frac{B_0}{4}$$

After the string breaks, the particles still move in circular path but now the radius is

$$r=\frac{mv}{qB}=\frac{m}{q(B_0/4)}\times\frac{qB_0L}{m}=4L$$

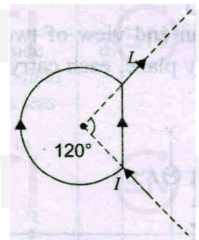
Let the particles but be at P and Q when the string breaks. At maximum seperation, they will be at R and S, as shown in figure.



$\therefore\ SR=4L+2L+2L+2L+4L=14L$

Paragraph for Question Nos. 25 to 26

In fig., the circular and the straight parts of the wire are made of same material but have different diameters. The magnetic field at the centre is zero.



25. Ratio of the currents I_1 and I_2 flowing through the circular and straight parts is

(a) $\frac{\sqrt{3}}{2\pi}$ (b) $\frac{2\sqrt{3}}{\pi}$

(c) $\frac{3\sqrt{3}}{2\pi}$ (d) $\frac{3\sqrt{3}}{2\sqrt{2}\pi}$

26. The ratio of the diameters of wires of circular and straight parts is

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{2\sqrt{3}}{\pi}$

(c) $\frac{3\sqrt{3}}{2\pi}$ (d) $\sqrt{2}$

Paragraph (Solution 25 to 26)

25.(c) The magnetic fields produced by I_1 in circular part and I_2 in straight part are equal and opposite. So,

$$\frac{2}{3}\times\frac{\mu_0I_1}{2R}=\frac{\mu_0I_2}{4\pi(R\cos60^o)}(\sin60^o+\sin60^o)$$

$$\therefore\ \frac{I_1}{I_2}=\frac{3\sqrt{3}}{2\pi}$$

26.(d) If R_1 and R_2 are the resistances of circular and straight parts respectively, then

$$I_1R_1=I_2R_2 \qquad \Rightarrow\ I_1\times\rho\frac{l_1}{A_1}=I_2\times\rho\frac{l_2}{A_2}$$

$$\Rightarrow\ \frac{A_1}{A_2}=\frac{I_1}{I_2}\times\frac{l_1}{l_2}=\frac{3\sqrt{3}}{2\pi}\times\frac{(4\pi/3)R}{2R\sin60^o}=2$$

$$\therefore\ \frac{d_1}{d_2}=\sqrt{\frac{A_1}{A_2}}=\sqrt{2}$$

Paragraph for Question Nos. 27 to 28

The current density in a linear conductor of radius a varies with r according to the relation $J=kr^2$, where k is a constant and r is the distance of a point from the axis of the conductor.

27. Find the magnetic field induction at a point at distance r from the axis when $r<a$. Assume relative permeability of the conductor to be unity.

(a) $\frac{\mu_0k\pi a^4}{4r}$ (b) $\frac{\mu_0kr^3}{2}$

(c) $\frac{\mu_0k\pi a^4}{2r}$ (d) $\frac{\mu_0kr^3}{4}$

28. Find the magnetic field induction at a point at distance r from the axis when $r > a$. Assume relative permeability of the medium surrounding the conductor to be unity.

(a) $\frac{\mu_0 k a^4}{4r}$ (b) $\frac{\mu_0 k r^3}{2}$

(c) $\frac{\mu_0 k \pi a^4}{2r}$ (d) $\frac{\mu_0 k r^4}{4}$

Paragraph (Solution 27 to 28)

27.(d) For $r < a$, the current through the cylinder of radius r is

$$I = \int_0^r J dA = \int_0^r kr^2 \times 2\pi r dr = 2\pi k \int_0^r r^3 dr = \frac{k\pi r^4}{2}$$

Using Ampere's law we have

$$B \times 2\pi r = \mu_0 \times \frac{k\pi r^4}{2} \qquad \therefore\ B = \frac{\mu_0 k r^3}{4}$$

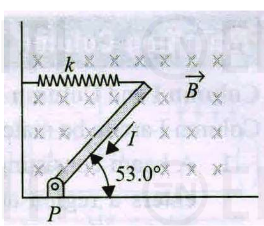
28.(a) The current through the complete conductor is

$$I_0 = \frac{k\pi a^4}{2}$$

For $r > a$, $B = \frac{\mu_0 I_0}{2\pi r} = \frac{\mu_0}{2\pi r} \times \frac{k\pi a^4}{2} = \frac{\mu_0 k a^4}{4r}$

Paragraph for Question Nos. 29 to 30

As shown in figure, a thin uniform rod of negligible mass and length 0.2 m is attached to the floor by a frictionless hinge at point P. A horizontal spring with force constant $k = 5\ \text{N m}^{-1}$ connects the other end of the rod to a vertical wall. The rod is in a uniform magnetic field $B = 2$ T directed into the plane of the figure.



There is current $I = 4$ A in the rod, in the direction shown.

29. The torque about P, due to the magnetic force on the rod is
(a) 0.16 Nm (b) 0.32 Nm
(c) 0.64 Nm (d) 0.28 Nm

30. When the rod is in equilibrium and makes an angle of 53° with the floor, the stretch in the spring is
(a) 1.6 m (b) 0.8 m
(c) 0.4 m (d) 0.2 m

Paragraph (Solution 29 to 30)

29.(a) Consider an element of the rod of length dr at distance r from P. The magnetic force on it is $dF = I\,Bdr$ and torque about P due to this force is $d\tau = rdF = IBrdr$

The total torque due to magnetic force is

$$\tau = IB\int_0^l rdr = \frac{IBl^2}{2} = \frac{4 \times 2 \times 0.2^2}{2} = 0.16\ \text{Nm}$$

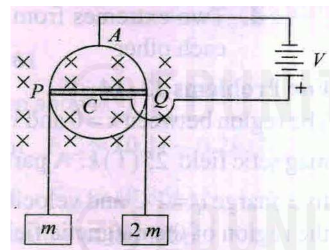
30.(d) In equilibrium, the torques about P due to spring force and magnetic force cancel each other.

$$\Rightarrow\ kx\,l \sin 53^\circ = \tau$$

$$\therefore\ x = \frac{\tau}{k\,l \sin 53^\circ} = \frac{0.16}{5 \times 0.2 \times 0.8} = 0.2\ \text{m}$$

Paragraph for Question Nos. 31 to 33

A conducting ring of mass m and radius r has a weightless conducting rod PQ of length $2r$ and resistance $2R$ attached to it along its diameter. It is pivoted at its center C with its plane vertical, and two blocks of mass m and $2m$ are suspended by means of a light inextensible string passing over it as shown in fig. The ring is free to rotate about C and the system is placed in a magnetic field B (into the plane of the ring). A circuit is now completed by connecting the ring at A and C to a battery of emf V. It is found that for certain value of V, the system remains static.



31. In static condition, the current through rod PC is
(a) V/R (b) $V/2R$
(c) $4V/R$ (d) $2V/R$

32. The value of V can be related with m, B and r as
(a) $2mgR / Br$ (b) mgR / Br
(c) $mgR / 2Br$ (d) $3mgR / Br$

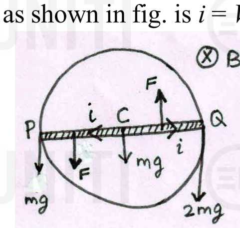
33. The net torque applied by the tension in strings is

(a) $\frac{3BVr^2}{R}$ (b) $\frac{BVr^2}{R}$

(c) $\frac{BVr^2}{3R}$ (d) $\frac{BVr^2}{2R}$

Paragraph (Solution 31 to 33)

31.(a) CP and CQ are in parallel to each other and p.d. across them is V. So, the current through each segment (CP and CQ) of the rod as shown in fig. is $i = V/R$.



32.(b) In equilibrium, the net torque about C must be zero.

$$\Rightarrow\ mgr + \frac{Fr}{2} + \frac{Fr}{2} = 2mgr \qquad \Rightarrow\ mg = F = iBr = \frac{V}{R}Br$$

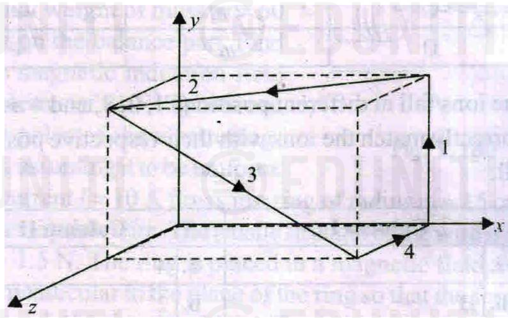
$$\therefore\ V = mgR / Br$$

33.(b) The net torque applied by the tension in strings is

$$2mgr - mgr = mgr = \left(\frac{V}{R}Br\right)r = \frac{BVr^2}{R}$$

Paragraph for Question Nos. 34 to 36

A wire carrying a 10 A current is bent to pass through various sides of a cube of side 10 cm as shown in fig. A magnetic field $\vec{B} = (2\hat{i} - 3\hat{j} + \hat{k})$ T is present in the region.



34. The net force on the loop is

(a) $\vec{F}_{net} = 0$ (b) $\vec{F}_{net} = (0.1\hat{i} - 0.2\hat{k})$ N
(c) $\vec{F}_{net} = (0.3\hat{i} + 0.4\hat{k})$ N (d) $\vec{F}_{net} = (0.36\hat{k})$ N

35. The magnetic moment vector of the loop is

(a) $(0.1\hat{i} + 0.05\hat{j} - 0.05\hat{k})$ Am^2
(b) $(0.1\hat{i} + 0.05\hat{j} + 0.05\hat{k})$ Am^2
(c) $(0.1\hat{i} - 0.05\hat{j} + 0.05\hat{k})$ Am^2
(d) $(0.1\hat{i} - 0.05\hat{j} - 0.05\hat{k})$ Am^2

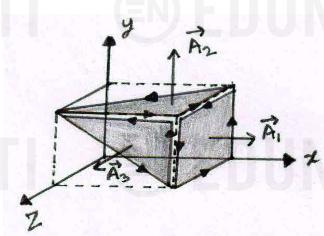
36. The net torque on the loop is

(a) $-0.1\hat{i} + 0.4\hat{k}$ Nm (b) $-0.1\hat{i} - 0.4\hat{k}$ Nm
(c) $0.2\hat{i} - 0.4\hat{k}$ Nm (d) $0.1\hat{i} - 0.4\hat{k}$ Nm

Paragraph (Solution 34 to 36)

34.(a) Since it is a closed loop and the magnetic field is uniform, the net force on the loop is zero.

35.(b) The given loop can be considered as a combination of three loops as shown in figure.



The area vectors of these three loops are

$\vec{A}_1 = (0.1 \times 0.1)\hat{i}$ m^2 $= 10^{-2}\hat{i}$ m^2,

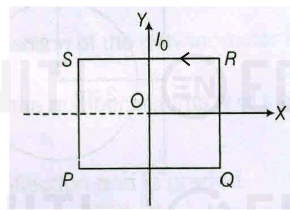
$\vec{A}_2 = \frac{10^{-2}}{2}\hat{j}$ m^2 and $\vec{A}_3 = \frac{10^{-2}}{2}\hat{k}$ m^2

$$\therefore \quad \vec{M} = i(\vec{A}_1 + \vec{A}_2 + \vec{A}_3) = 10\left(10^{-2}\hat{i} + \frac{10^{-2}}{2}\hat{j} + \frac{10^{-2}}{2}\hat{k}\right)$$
$$= (0.1\hat{i} + 0.05\hat{j} + 0.05\hat{k})\,\text{Am}^2$$

36.(c) $\vec{\tau} = \vec{M} \times \vec{B} = (0.1\hat{i} + 0.05\hat{j} + 0.05\hat{k}) \times (2\hat{i} - 3\hat{j} + \hat{k})$
$= (0.2\hat{i} - 0.4\hat{k})$ Nm

Paragraph for Question Nos. 37 to 38

A uniform, constant magnetic field B_0 is directed at an angle of 45° to x– axis in xy-plane. $PQRS$ is a rigid square wire frame carrying a steady current I_0 with its centre at origin O. At time $t = 0$, the frame is at rest in the position shown in the figure with its side parallel to x and y-axis. Each wire of the frame has mass M and length L.



37. The torque acting on the frame due to magnetic field and axis about which it will rotate is

(a) $\frac{I_0L^2B_0}{\sqrt{2}}(\hat{i} + \hat{j})$ and SQ (b) $\frac{I_0L^2B_0}{\sqrt{2}}(-\hat{i} + \hat{j})$ and SQ
(c) $\frac{I_0L^2B_0}{2}(-\hat{i} - \hat{j})$ and PR (d) $\frac{I_0L^2B_0}{2}(\hat{i} + \hat{j})$ and PR

38. Find the angle by which the frame rotates under the action of this torque in a short interval of time Δt. (Δt is so short that any variation in the torque during this interval may be neglected)

(a) $\frac{3I_0B_0}{4M}(\Delta t)^2$ (b) $\frac{4I_0B_0}{M}(\Delta t)^2$
(c) $\frac{2I_0B_0}{3M}(\Delta t)^2$ (d) $\frac{4I_0B_0}{3M}(\Delta t)^2$

Paragraph (Solution 37 to 38)

37.(b) Here, $\vec{M} = I_0L^2(\hat{k})$ and $\vec{B} = \frac{B_0}{\sqrt{2}}(\hat{i} + \hat{j})$

$$\therefore \vec{\tau} = \vec{M} \times \vec{B} = I_0L^2(\hat{k}) \times \frac{B_0}{\sqrt{2}}(\hat{i} + \hat{j}) = \frac{I_0L^2B_0}{\sqrt{2}}(-\hat{i} + \hat{j})$$

It will rotate about axis SQ.

38.(a) On collapsing the frame into a single straight wire PR, we now have a uniform wire of mass $4M$ and length $\sqrt{2}L$. Its MOI about axis SQ is

$$I = \frac{(4M)(\sqrt{2}L)^2}{12} = \frac{2ML^2}{3}$$

In short interval of time Δt, as the torque is considered constant, the angle $\Delta\theta$ by which the frame rotates is

$$\Delta\theta = \frac{1}{2}\alpha(\Delta t)^2 \quad \text{where } \alpha = \frac{\tau}{I} = \frac{I_0 L^2 B_0}{2ML^2/3} = \frac{3I_0 B_0}{2M}$$

$$\therefore\ \Delta\theta = \frac{1}{2}\times\frac{3I_0 B_0}{2M}\times(\Delta t)^2 = \frac{3I_0 B_0}{2M}(\Delta t)^2$$